

Afterbloom

Daily Check-in Feature Specification

Version 1.0 | Developer Brief | Postpartum Depression Monitoring App

Flow input

Step 0: Welcome Screen

จุดประสงค์: บอกแม่ว่าแอปนี้ไม่ใช่แอฟวินิจฉัยโรค แต่เป็นเพื่อนช่วยติดตามหลังคลอด
Screen copy

Welcome to AfterBloom A gentle space to check in with yourself after birth.
Button: Start my care journey
หน้านี้ยังไม่ต้องเก็บ input

Step 1: Basic Setup Input

อันนี้คือกรอกครั้งแรกเท่านั้น เพื่อให้ระบบรู้ว่าแม่อยู่ stage ไหน

1.1 Mother name / nickname

คำถาม: อยากให้ AfterBloom เรียกคุณว่าอะไร?
Input type: text field Example: Sia / Moji / แม่เมย์
Developer field: mother_name

เอาไปใช้: แสดง greeting บน Home เช่น “Good morning, Sia”

จำเป็น**

1.2 Baby’s birth date

คำถาม: ลูกเกิดวันที่เท่าไร?
Input type: date picker
Developer field: baby_birth_date
เอาไปใช้: ระบบคำนวณเองว่าแม่อยู่ Day / Week / Month ไหน เช่น Today - baby_birth_date = postpartum_day postpartum_day 14 = Week 2

postpartum_day 42 = Week 6 postpartum_day 90 = Month 3
สำคัญมาก: อันนี้คือ input ที่สำคัญที่สุดใน onboarding เพราะมันทำให้ Home แสดง stage badge และ Care Journey ได้
ห้ามตัด

1.3 First-time mother?

คำถาม: นี่เป็นลูกคนแรกของคุณไหม?
Choices:
- ใช่ ลูกคนแรก - ไม่ใช่ - ไม่อยากตอบ
Developer field: is_first_time_mother
เอาไปใช้: ปรับ tone ของคำแนะนำ เช่น แม่ลูกคนแรกอาจต้องการ guidance ที่อธิบายละเอียดกว่า
จำเป็นไหม: ควรมี แต่ถ้า prototype ต่วนมาก ตัดได้

1.4 Delivery hospital

คำถาม: คุณคลอดที่โรงพยาบาลไหน?
Input type: dropdown / text field
Developer field: hospital_name
เอาไปใช้: ทำให้ model B2B ดูดีขึ้น เช่น Home อาจขึ้นว่า "Connected with KKU Hospital Care Journey"
จำเป็นไหม: ถ้า pitch เน้น hospital integration ควรมี

1.5 Preferred check-in time

คำถาม: อยากให้เราเตือน check-in เวลาไหน?
Choices:
- Morning - Afternoon - Evening - Before bed - I'll check in by myself
Developer field: preferred_checkin_time

เอาไปใช้:

notification setting / reminder mockup

สำคัญ:

ใน research ที่โมจิใส่ไว้ แม่อยากควบคุมเวลาแจ้งเตือนเอง ไม่ใช่ให้แอปบังคับเวลา

1.6 Privacy preference

คำถาม:

ถ้าวันไหนคุณต้องการ support เพิ่ม อยากให้ใครเห็นข้อมูลนี้ไหม?

Choices:

- ยังไม่แชร์กับใคร
- แชร์กับคนในครอบครัวที่ฉันเลือก
- แชร์กับ care team เท่านั้น
- ค่อยตั้งค่าทีหลัง

Developer field:

sharing_preference

เอาไปใช้:

Invite Your Circle / hospital dashboard phase 2

สำหรับ prototype:

ทำเป็นหน้า mock ได้ ยังไม่ต้องทำระบบแชร์จริง

สรุป Onboarding Input ที่ต้องมีจริง

1.mother_name

2.baby_birth_date

3.First-time mother?

4.preferred_checkin_time

ถ้าเวลาน้อยมาก ๆ เหลือแค่ 2:

baby_birth_date

preferred_checkin_time

Step 2: Home Screen Output

หน้า Home ไม่ควรเป็นหน้ากรอกข้อมูลเยอะ

Home คือหน้าที่ “ดึง input มาแสดง”

Home ควรขึ้นหลัง onboarding แบบนี้

***สีเหลืองคือข้อความ

***สีเขียวคือปุ่ม หรือ สิ่งที่ไขว่

Good morning, Sia

You are in Week 2 after birth

How are you feeling today?

[Start 30-sec check-in]

Today's care journey:

Week 2 is a sensitive emotional adjustment period.

Your mood this week

[Mood Trend 7 วัน]

Tiny goal:

Take one deep breath before feeding

[I Need Help]

Home ใช้ข้อมูลจาก input ไหน

Home element	มาจาก input
Greeting	mother_name
Stage badge	baby_birth_date
Check-in CTA	daily check-in status
Care Journey	postpartum stage
Support Level	daily check-in answers
Mood Trend	check-in history
I Need Help	always visible

ในไฟล์ของโมจิเองก็วางไว้ว่า Home ควร มี Contextual greeting, Check-in CTA, Care Journey, Support Level, Help button, Tiny Goal และ Mood Trend โดยทุก element มีเหตุผลรองรับ ไม่ใช่เลือกตาม preference เฉย ๆ

Section 3: Daily Check-in Input Specification

The daily check-in is the core touchpoint of the Afterbloom app. It collects 4 validated clinical indicators every day, with conditional follow-up questions triggered by low scores. All questions use closed-ended 1–5 scales to minimize cognitive load for postpartum mothers.

Design principle: One question per screen. Tap to answer. Target completion time under 60 seconds for the core 4 indicators.

Phase 1: Core Daily Check-in (Always Asked)

The following 4 indicators are asked every day, in this order. All use a 1–5 scale where 1 = worst, 5 = best, EXCEPT anxiety_score which is reversed (1 = not at all, 5 = overwhelmed).

3.1 Mood

Question	How are you feeling today?
UI format	Emoji scale / card buttons — one per screen

Developer field	mood_score
Data type	number 1–5 (1 = worst, 5 = best)

Choices:

1	 Very low
2	 Low
3	 Neutral
4	 Good
5	 Very good

📌 *Research basis: EPDS item 1 (able to laugh / see funny side) + PHQ-9 item 1 (depressed mood). EMA pilot PMC 2022 (n=26): self-reported mood has predictive validity against EPDS follow-up score.*

3.2 Sleep

Question	How did you sleep last night? (even if you woke up often)
UI format	Card buttons — one per screen
Developer field	sleep_score
Data type	number 1–5 (1 = worst, 5 = best)

Choices:

1	Almost none
2	Very little
3	Some
4	Enough
5	Slept well

📌 *Note: The phrase 'even if you woke up often' is intentional — postpartum mothers wake frequently for feeding. Framing removes guilt about broken sleep and improves response accuracy.*

📌 *Research basis: EPDS item 7 (difficulty sleeping). Okun et al. 2018 (SLEEP): sleep disturbance is a prodromal symptom appearing weeks before full PPD onset. EMA pilot PMC 2022: sleep quality has predictive validity against EPDS score.*

3.3 Energy

Question	How much energy do you have today?
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UI format	Card buttons — one per screen
Developer field	energy_score
Data type	number 1–5 (1 = worst, 5 = best)

Choices:

1	Drained
2	Tired
3	Managing
4	Okay
5	Energized

📌 *Research basis: DSM-5 criterion — fatigue or loss of energy as core MDD symptom. PubMed 2012 brief scales: 'slowed down' / psychomotor retardation is a top predictor of major depressive episode postpartum. Russell 1980 circumplex model: activation is the second axis after valence.*

3.4 Worry

Question	How much is on your mind today?
UI format	Card buttons — one per screen
Developer field	worry_score
Data type	number 1–5 (REVERSED: 1 = not at all, 5 = overwhelmed)

Choices:

1	Not at all
2	A little
3	Somewhat
4	A lot
5	Overwhelmed

📌 *IMPORTANT for developer: worry_score is REVERSED. A score of 5 = overwhelmed = worst state. When computing composite score, invert this field: adjusted_worry = 6 - worry_score. Do not store the inverted value — store raw input, invert during scoring.*

📌 *Research basis: Goodman 2009 (JOGNN): ~50% of mothers with PPD have comorbid anxiety disorder. EMA pilot 2022: anxiety has predictive validity against EPDS. Cognitive worry / rumination is the dominant anxiety presentation in PPD, not physiological panic.*

Phase 2: Conditional Follow-up (Triggered When Output Is Not Good)

These questions appear only when the composite score from Phase 1 falls below threshold. They do not appear on every check-in.

Trigger condition	composite_score < 2.5 OR any single core indicator score = 1
Phase 2 flow order	3.5 Problem Tag → 3.6 Optional Journal → 3.7 Baby Connection

3.5 Problem Tag

Question	What feels hardest today?
UI format	Multi-select tag chips
Developer field	problem_tags
Data type	array of enum strings maximum 2 selections
Additional field	problem_other_text text / nullable max 150 characters
Required	problem_other_text required only if 'Something else' is selected

Choices (select up to 2):

1	Exhaustion
2	Feeding
3	Baby crying
4	Body pain
5	Sleep deprivation
6	Feeling alone
7	Family stress
8	Partner support
9	Recovery after birth
10	Household responsibilities
11	Feeling overwhelmed
12	I'm not sure
13	Something else

🚩 If 'Something else' is selected: show short text field. Placeholder: 'Tell us what feels hardest today.'

3.6 Optional Journal

Question	Would you like to write a little more about today?
Description shown	This is optional. You can leave it blank.
Input type	Text area
Placeholder	Write anything you want to remember, release, or explain about today.
Developer field	journal_entry
Data type	text / nullable
Character limit	500 characters
Required	No — must never block check-in flow. Mother can skip anytime.

3.7 Baby Connection

Question	How did moments with your baby feel today?
UI format	Card buttons — one per screen
Developer field	baby_connection_score
Data type	number 1–5 (1 = worst, 5 = best)

Choices:

1	Hard
2	A little distant
3	Mixed
4	Mostly okay
5	Warm

🚩 Important: Framing must not imply judgment. 'A little distant' should not feel like a failure label. Avoid words like 'disconnected' or 'bad mother'. The goal is honest reporting, not guilt.

🚩 Research basis: Postpartum Bonding Questionnaire (Brockington 2001). EPDS bonding construct. Poor bonding is associated with downstream child developmental risk (Murray 1992, Br J Psychiatry).

Phase 3: Support Need (Triggered by Pattern, Not Every Check-in)

This question is shown only when a specific pattern is detected. It is NOT part of the daily conditional flow. It must not appear more than once per unresolved episode.

3.9 Support Need

Question	Do you need our hospital team to reach out to you today?
UI format	Two large tap buttons
Developer field	support_request
Data type	boolean true = Yes please false = Not right now

Choices:

1	Yes, please
2	Not right now

Trigger logic:

**** สำหรับ Single day crash (จุ๊ทุกอย่างแยลงหนักมากภายในวันเดียว) หรือ ถ้าหาก mood trend ต่ำรวม 3 วัน**

Trigger A	composite_score < 2.5 for 3 consecutive days
Trigger B	mood_score = 1 AND worry_score = 5 on the same day (single day crash)
After 'Yes, please'	support_request = true is sent to hospital dashboard. Question disappears until hospital marks episode as resolved.
After 'Not right now'	Wait 3 days. Re-trigger if scores remain low.

► *Rationale: Asking this question every day a mother feels low causes habituation (automatic 'no' responses) and increases burden. Triggering on pattern ensures the question is timely and meaningful. EMA pilot 2022 supports 3-day trend as a clinically significant threshold.*

Always Accessible: I Need Help Button

The I Need Help button is always visible on the home screen and does not require a check-in to access. This replaces a daily Safety Check question in the check-in flow.

Location	Home screen — persistent button, always visible
Purpose	Safety check crisis access EPDS item 10 equivalent
Action on tap	Route to: emergency contact / hospital contact / 'I am with someone safe'

Developer field	safety_access_used
Data type	boolean / timestamp log each use

- ▀ *Clinical basis: ACOG 2023 mandatory safety screening. npj Digital Medicine 2023: crisis access must be 1-tap, must not require completion of other steps. EPDS item 10 (thoughts of self-harm) is the only item requiring immediate action regardless of total score.*
- ▀ *Rationale for not asking daily: Daily safety questions cause sensitization — mothers begin answering automatically without genuine reflection, reducing clinical validity (Shiffman et al. 2008).*

Section 4: Check-in Result Output

After completing the daily check-in, the app displays a result screen immediately. The result communicates the mother's Support Level for the day.

Language requirement: Do NOT use the term 'Risk Level'. Use 'Today's Support Level'.
Diagnostic labels and clinical risk language increase stigma and reduce engagement. This is supported by the app's own survey data (n=320, 72% feared judgment).

4.1 Scoring Logic

mood_score	Raw score 1–5 (1 = worst)
sleep_score	Raw score 1–5 (1 = worst)
energy_score	Raw score 1–5 (1 = worst)
worry_score (inverted)	adjusted_worry = 6 - worry_score (so 1 = best, 5 = worst)
composite_score	(mood_score + sleep_score + energy_score + adjusted_worry) / 4
composite range	1.0 = all worst 5.0 = all best
Override rule	If support_request was triggered and baby_connection_score = 1: escalate to Level 3 regardless of composite

4.2 Support Level Definitions

Level 1 Steady	
Trigger condition	composite_score >= 3.5
Message to mother	<i>You seem steady today. Keep going at your own pace.</i>
CTA buttons	→ View today's care tip → See your mood trend
Research basis	Positive reinforcement increases sustained engagement (JMIR Human Factors, MBapp 2024). Steady days should feel rewarding, not neutral.

Level 2 Gentle Support	
Trigger condition	composite_score >= 2.5 AND < 3.5

Message to mother	<i>Today might feel a little heavy. You don't have to push through everything at once.</i>
CTA buttons	→ Choose one small goal → Breathing exercise
Research basis	Behavioral activation (CBT micro-actions) for mild symptoms has evidence in MBapp 2024. Small achievable actions reduce learned helplessness.

Level 3 Extra Support Recommended

Trigger condition	composite_score >= 1.5 AND < 2.5 OR mood_score <= 2 AND adjusted_worry >= 4
Message to mother	<i>You may need a little more support right now. Afterbloom recommends an emotional check-in or talking to someone you trust.</i>
CTA buttons	→ Take emotional screening (EPDS) → Talk to your support circle → Contact your hospital team
Research basis	EMA pilot 2022: 3-day low trend has predictive validity for EPDS high score. Level 3 triggers EPDS screening prompt.

Level 4 Immediate Support

Trigger condition	support_request = true (mother tapped 'Yes, please' on Support Need question)
Message to mother	<i>You don't have to go through this alone. We'll connect you with support right away.</i>
CTA buttons	→ Contact hospital team → Call emergency line → I'm with someone safe
Research basis	Triggered by mother's explicit request for help. Hospital dashboard receives alert. ACOG 2023: hospital follow-up is mandatory when support need is flagged.

4.3 Hospital Dashboard Alerts

Alert A	support_request = true → immediate notification to hospital staff dashboard
Alert B	composite_score < 2.5 for 3 consecutive days → flag on hospital dashboard (no immediate notification required)
Alert C	safety_access_used = true → log and flag on hospital dashboard

Resolution	Hospital staff marks episode as resolved in dashboard. support_request question re-enables after resolution.
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► *Integration note: Hospital dashboard integration is required for Alert A to function. Confirm HIS integration scope and data format with pilot hospital before development.*

เพิ่มเติม

4.1 Addendum — Scoring Logic (Edge Cases)

Partial check-in: If the mother exits before completing all 4 core indicators, composite_score is not calculated for that day. Previously entered responses are saved and remain visible when the mother reopens the app. The check-in session remains resumable until 11:59 PM of the same day.

Score weighting: All four core indicators carry equal weight (25% each). Formula: $\text{composite_score} = (\text{mood_score} + \text{sleep_score} + \text{energy_score} + \text{adjusted_worry}) / 4$. Weighting will be reviewed and calibrated during the pilot phase based on clinical outcomes data.

4.3 Addendum — Hospital Dashboard Data

When support_request = true or composite_score < 2.5 for 3 consecutive days, the hospital dashboard receives: mother ID, alert type, composite_score, and all four individual scores (mood, sleep, energy, worry). Rationale: clinical staff require indicator-level context to conduct a meaningful follow-up call. Sending a flag only increases time-to-support.

Data transmitted follows minimum necessary principle per PDPA Thailand requirements.

4.4 Contact Hospital — Available Options (Level 3 & Level 4)

When the mother taps "Contact your hospital team", two options are presented:

Option 1 — Call now: Displays the hospital's direct phone number. Tapping dials immediately. App logs the action as hospital_contact_initiated.

Option 2 — Request callback: App sends a notification to the hospital dashboard indicating the mother has requested contact. Hospital staff follow up by phone. App displays confirmation: "Your hospital team has been notified and will reach out to you soon."

Note: Option 2 requires hospital dashboard integration to function. Confirm availability with pilot hospital before development. Option 1 is available as a fallback regardless of integration status.

Key references: DSM-5, EPDS (Cox et al. 1987), EMA Pilot PMC 2022, Okun et al. 2018 (SLEEP), Goodman 2009 (JOGNN), ACOG 2023, npj Digital Medicine 2023, MBapp 2024 (JMIR Human Factors), PubMed 2012 brief scales, Russell 1980 circumplex model, Brockington 2001 (PBQ)

Need Update ยังไม่ confirm

Step 5: Tiny Goal Input / Output

หลัง result ให้แม่เลือก goal ได้ 1 อย่าง ไม่ต้องเยอะ

คำถาม

วันนี้อยากลองทำสิ่งเล็ก ๆ อะไรหนึ่งอย่าง?

Choices:

- ดื่มน้ำหนึ่งแก้ว
- ส่งข้อความหาคนที่ไวใจ
- กลับตาทายใจลึก 30 วินาที
- วางมือบนอกแล้วบอกตัวเองว่า “ฉันกำลังพยายามอยู่”
- ขำวันนี้ก่อน

Developer field:

selected_tiny_goal

Data type: enum/string

สำคัญ:

ต้องมี “ขำวันนี้ก่อน” เพราะแม่บางวันไม่ไหวจริง ๆ และในไฟล์โมจิก็มี research support เรื่อง defer/snooze option ว่าช่วยเรื่อง acceptability

Step 6: Care Journey Input

Care Journey จริง ๆ ไม่ต้องให้แม่กรอกเพิ่มเยอะ

มันควรใช้ baby_birth_date เพื่อ generate stage อัตโนมัติ

Stage mapping

Day 0–3 = First Days

Day 4–13 = Early Adjustment

Day 14–21 = Week 2 Checkpoint

Day 22–42 = First 6 Weeks

Month 2–3 = Settling Phase

Month 4–6 = Returning Rhythm

Month 7–9 = Late Emotional Check

Month 10–12 = First Year Reflection

Care Journey card บน Home

ถ้าแม่อยู่ Week 2:

Week 2 Checkpoint

Many mothers feel emotionally sensitive during this period.

If sadness, numbness, or anxiety continues, a short emotional check can help you understand what support you may need.

CTA:

Start Week 2 emotional check

ในไฟล์โมจิ support ว่า Care Journey ไม่ควรถูกซ่อนใน tab เพราะแม่ต้องการ emotional change info บน Home และช่วง Week 2 / Week 6 เป็นช่วงสำคัญของ PPD

Step 7: EPDS-based Screening Input

อันนี้ไม่ใช่ daily input

ให้ขึ้นเฉพาะบางจังหวะ เช่น Week 2 / Week 6 / Month 3 / Month 6 / Month 9 / Month 12

Trigger

EPDS-based screening ควรขึ้นเมื่อ:

postpartum_stage = Week 2
or postpartum_stage = Week 6
or support_level = Extra Support Recommended
or safety_status = some
or mood_score <= 2 for 3 days
or anxiety_score >= 4 for 3 days

Wording หน้าแรก

ไม่ใช่คำว่า:

Depression Test

ใช้ว่า:

Emotional Check

คำถามสั้น ๆ เพื่อดูว่าช่วงนี้คุณต้องการ support แบบไหน

Input type

ใช้คำถามแบบ EPDS-based แต่ใน prototype ไม่ต้องใส่ครบทุก clinical scoring ก็ได้ ถ้ากลัวผิด ให้ทำเป็น mock flow ก่อน

Developer field:

epds_answers[]

epds_total_score

screening_date

screening_trigger

Output:

screening_result_level

recommended_next_step

Step 8: Help Input / Emergency Flow

ปุ่ม I Need Help ต้องอยู่ทุกหน้า ไม่ใช่แค่หน้า result

เมื่อกด I Need Help

ให้ถามแค่ 1 ข้อ:

ตอนนี้คุณต้องการความช่วยเหลือแบบไหน?

Choices:

- อยากคุยกับคนที่ไว้ใจ
- อยากติดต่อโรงพยาบาล
- อยากอ่านวิธีรับมือเบื้องต้น
- รู้สึกไม่ปลอดภัยกับตัวเอง

ถ้าเลือก “รู้สึกไม่ปลอดภัยกับตัวเอง”

ไปหน้า urgent support ทันที

You deserve immediate support.

Please contact emergency help or stay with someone you trust now.

Prototype ยังไม่ต้องโทรจริง แต่ควรมีปุ่ม mock:

- Call emergency support
- Contact hospital
- Message trusted person

ในไฟล์โมจิระบุว่า Help access ต้อง visible ทุก state เพราะ distress ไม่ได้เกิดเฉพาะเวลาทำการ และแม่อาจใช้แอปตอนกลางคืน/ตี 4

